

FAN-XUE GABRIELLA REIDYS

Website: gabriellaxue.github.io

Email: fanxue2@illinois.edu

RESEARCH INTERESTS

Satellite Networks, Traffic Engineering, Scalability and Efficiency of Networked Systems, Datacenter Networks, Internet of Things

EDUCATION

PhD in Computer Science, University of Illinois at Urbana-Champaign *Aug. 2023 - Present*

Advised by Professor Matthew Caesar

MS in Computer Science, University of Illinois at Urbana-Champaign *Aug. 2023 - Aug. 2024*

Masters Thesis: *A Metaverse for the Internet of Things*

Advised by Professor Matthew Caesar

BS in Computer Science, University of Illinois at Urbana-Champaign *Aug. 2018 - May 2023*

High Honor

PUBLICATIONS

Vivisecting Starlink Throughput: Measurement and Prediction. Zikun Liu, Fan-Xue Gabriella Reidys, Sarah Tanveer, Deepak Vasisht. *Proceedings of the 21st International Conference on Emerging Networking Experiments and Technologies (CoNEXT'25)*.

Towards Immersive Cloud-Based IoT Education. Fan Gabriella Xue, Matthew Caesar. *ACM SIGCOMM Computer Communication Review (CCR 2024)*.

RESEARCH IMPACT

Optimization of Satellite Networking

- We identified that satellite networks suffer from large bandwidth fluctuations over short time scales due to several factors, including satellite mobility and weather. These fluctuations can lead to worse application performance, with users around the world reporting frequent network disconnection and unstable network. To address this, we propose a learning-based throughput prediction system that leverages satellite networks' unique characteristics. Our system improves throughput-prediction accuracy by 30% and can be used to improve video streaming QoE by 2X.
- As satellite networks grow rapidly, we also aim to redesign traffic engineering and congestion control systems for satellite networking.

Scaling Policy Enforcement in Hyperscale Cloud Gateways

- Cloud gateways sit on the critical forwarding path between tenants, services, and external networks, where they enforce both routing and security policies at line rate. This requires efficient data-plane support for packet lookup tasks. As cloud-scale policy tables outgrow the limited on-chip SRAM of programmable switches, recent systems increasingly place large classification tables in off-chip high-bandwidth memory, which makes memory traffic a primary bottleneck. We aim to co-design the group function and hash functions to reduce classification to a small, bounded sequence of exact-match queries to improve latency for packet classification.

IoT Virtualization Platform

- An increasing number of students are becoming interested in learning about the IoT space. However, today we lack scalable and efficient ways to bring hands-on IoT learning to many due to hardware accessibility, system complexity, and deployment environment constraints. Therefore, we propose the architecture of a virtual shared space where people can design IoT systems, create virtual worlds, and deploy systems within these worlds to observe and engage with these systems. We evaluated our system's usability through statistical analyses of 355 students, resulting in positive learning outcomes.
- The current system is deployed to more than 600 students from multiple universities worldwide, as well as tens of high schools.

WORK EXPERIENCE

Bytedance

Network R&D Intern

May 2025 - August 2025

Bellevue, WA

- Worked on handling hyperscale traffic for the internet gateway in the data center network.
- Set up ByteDance's virtual networking infrastructure in a customized test cluster to collect performance metrics.

Accelerated Learning and Engineering Research Training Program

Research Mentor

May 2023 - Dec. 2023

Urbana, IL

- Advised two undergraduate students on research in IoT systems, guiding them in methodology and experiment design, results analysis, and fostering team collaboration.

C3.ai

Software Engineering Intern

May 2023 - Aug. 2023

Redwood City, CA

- Designed, implemented, and tested anomaly detection components in the sourcing optimization application from end to end.

C3.ai

Software Engineering Intern

May 2022 - Aug. 2022

Redwood City, CA

- Built the ESG app CSO persona from scratch that unifies fragmented ESG data, identifies priority issues, tracks progress against goals, and automates reporting aligned to key standards.

Marqeta

Software Engineering Intern

June 2021 - July 2021

Oakland, CA

- Built a NACHA file viewer endpoint to decrypt/encrypt and parse information for both technical and non-technical users to verify/download transaction content in a human-readable format securely, and simplified SFTP uploading via endpoints.

TEACHING EXPERIENCE

CS423 Operating System Teaching Assistant

Spring 2026

- Taught a guest lecture about networking component in operating system.
- Held office hours, answered questions, and developed assignments related to operating systems.

CS425 Distributed System Teaching Assistant

Fall 2025

- Held office hours, answered questions, and developed assignments related to distributed systems.

CS437 Intro to IoT Teaching Assistant

Spring 2023 & 2025

- Held office hours and answered questions on topics related to the Internet of Things.
- Managed the deployment and maintenance of the virtual lab infrastructure for 200 students.

CS341 System Programming Teaching Assistant

Fall 2024

- Lectured weekly lab sections and answered questions on system programming.

CS225 Data Structure Teaching Assistant

Fall 2022

- Held office hours and lab sections to teach and answer questions on data structures.
- Named to the *List of Teachers Ranked as Excellent by Their Students*.

AWARDS AND ACTIVITIES

- CoNEXT Travel Grant (2025)
- NSDI Travel Grant (2025)
- Sysnet Woman Student Organizer (2024)
- List of Teachers Ranked as Excellent by Their Students (2022)
- Usenix Student Grant Program (OSDI 2022)
- Fall SoS Quarterly Lablet Meeting Student Contributor (2021)
- Yunni and Maxine Pao Memorial Scholarship (2021-2022)
- TBP Outstanding Junior Awards (2021)
- Student volunteer in SIGCOMM (2021)
- Illinois Engineering Achievement Scholarship (2021)
- James Scholar (2018-2022)

PROGRAMMING LANGUAGES

- **Advanced: Python**
- **Intermediate: Java, C++, C, JavaScript, HTML, CSS, SQL**